

REUBEN SAMSON RAJ

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[GitHub](#) • [Linkedin](#) • [Google Scholar](#)

CAREER SUMMARY

Systems and security engineer with 9+ years in telecom-network infrastructure and Ph.D. research in critical infrastructure security. Experienced in cyber-physical security, networking, cloud-native software deployment, programmable data planes, software engineering, test automation and 3GPP specifications.

EDUCATION

Ph.D. in *Computer Science*, University of Arkansas, Fayetteville, AR Aug 2021 – Dec 2026 (Expected)

M.S. in *Computer Engineering*, University of Arkansas, Fayetteville, AR Aug 2021 – Dec 2024

B.E. in *Electronics and Communication Engineering*, College of Engineering, Guindy, Anna University, Chennai Aug 2006 – June 2010

SKILLS

Languages	: Python, P4(Tofino, BMv2), C, Bash
Networking	: SDN, P4, EPC/5G Core, TCP/IP, Mininet, Wireshark, iperf
Cloud & Orchestration	: Kubernetes, Docker, OpenStack
Security	: IPSec, TLS, ICS/OT (Modbus, DNP3) Network Security, IDA Pro, PE Studio
Tools	: Linux, Git, FastAPI, OpenCV, NVIDIA GPU Stack
Scripting/Automation	: TCL, Robot Framework, Cucumber

WORK EXPERIENCE

University of Arkansas, Fayetteville, AR

Senior Graduate Assistant Aug 2021 – Present

- Independently research and publish on secure Smart Grid/ICS networks using P4-programmable switches
- Act as the go-to person for issues, including Tofino setup, coding and advising lab peers on technical solutions
- Mentor and provide research directions for undergrads on thesis projects
- Recognized by advisor for research independence, peer-level contributions, and leadership

Nokia Bell Labs, New Providence, NJ

Network Architecture Intern Jun 2023 – Aug 2023

- Deployed Kubernetes clusters on Nokia Airframe servers, simulating multi-cloud setups
- Optimized YoLov7-based object detection by transforming it into a FastAPI REST service, reducing RTT latency from 500 ms to 30 ms
- Integrated NVIDIA GPU plug-ins with Kubernetes to enhance GPU utilization for inference pods
- Scaled the video inference system, enabling real-time webcam feed processing with OpenCV

Cisco Systems, Bangalore

Software Engineer Jun 2020 – Jul 2021

- Integration and end-to-end testing for Cisco's Cloud-Native 5G Core network functions – SMF, UPF, SGW
- Deployed and managed Kubernetes-based software deployments in production environments
- Uncovered several critical bugs on RADIUS interface and led automation strategy for the same

Parallel Wireless, Bangalore

Member of Technical Staff Oct 2018 – May 2020

- Performed QA validation for Parallel Wireless' gateway products – TWAG (WiFi gateway) and SAEGW
- Validated VoWiFi call scenarios over GRE/IPsec tunnels and verified IKEv2 tunnel establishment
- Developed Python automation for SAEGW Gx-interface tests; reduced manual test execution time by ~60%

Nokia Networks, Bangalore
R & D Engineer

Apr 2015 – Oct 2018

- Software validation for Nokia's Home Subscriber Server (HSS) and 5G UDM Network Function
- Hands on experience with Virtual Network Functions (VNF) bring up on OpenStack, Nokia CBAM (Cloud Band Application Manager)
- Led test strategy for HSS 5G VNF, improving defect detection before customer deployments and reducing field issues

Cisco Systems, Bangalore
Software Engineer

Apr 2012 – Apr 2015

- Performed software verification on Cisco StarOS ASR 5000, ASR 5500 SGSN gateway nodes
- Developed and maintained automated regression test suites for SGSN features using TCL scripting, enhancing Cisco's test automation framework for improved efficiency

Accenture, Chennai
Associate Software Engineer

Apr 2011 – Apr 2012

- Performed software testing for a variety of Credit Card delinquency applications for Bank of America

SELECTED PUBLICATIONS (full list [here](#))

1. **Reuben Samson Raj** and Dong Jin. *A Framework to Evaluate PMU Networks for Resiliency Under Network Failure Conditions*, IEEE SmartGridComm 2022
2. **Reuben Samson Raj** and Dong Jin. *Dynamic Data Driven Security Framework for Industrial Control Networks using Programmable Switches*. DDDAS 2024
3. **Reuben Samson Raj** and Dong Jin. *Leveraging Data Plane Programmability Towards a Policy-driven In-Network Security Framework for Industrial Control Systems*. ACM e-Energy 2025
4. Sarvesh Bidkar, Rakesh Abbireddy, Maryam Amiri, **Reuben Samson Raj**, Anil Rana, Jeff McLaird, Paul Rea, Meryem Simsek and Jesse E. Simsarian. *Demonstration of Multi-Provider Network and Cloud Service Provisioning with Blockchain Smart Contracts*. ECOC 2025
5. **Reuben Samson Raj**, Dong Jin. *A Dynamic Data-Driven Framework for Enhancing Energy System Security and Resilience through Programmable Networks*. Book Chapter - DDDAS Handbook Vol. IV, Springer (Under Production)
6. Zhiyao He*, **Reuben Samson Raj***, Yanfeng Qu, Dong Jin. *Leveraging Programmable Switches towards Machine Learning-based Anomalous Event Classification of PMU data*. IEEE ICC 2026 (* - Equal Contribution)
7. Yuexin Chen, **Reuben Samson Raj**, Xiaoyu Guo, Dong Jin. *Enhancing Security of Grid Control Systems through In-network Collaboration of Programmable Data Planes*. IEEE SmartGridComm 2026 (Accepted)

AWARDS & HONORS

- **2nd place** - Best Paper Award, DDDAS 2024
- Reginald R. Barney & Jameson A. Baxter Graduate Scholarship (2023, 2022)
- Department of Electrical Engineering and Computer Science Scholarship (2023)
- You Amaze and You Inspire awards (2020), CAP Award (2013), Cisco
- Best Feature Team Award (2016), Nokia

ACADEMIC SERVICE & VOLUNTEERING

- Journal Reviewer, ACM TOMACS, 2025, 2026
- Conference Reviewer, IEEE SmartGridComm, 2026
- Key organizing member for high-school programming contest; technical liaison with coding platform vendor testing and onboarding group of volunteer judges